PROBABILITY AND STATISTICS

Dr. Öğr. Üyesi Sadra Mousavi

1st semester 2024-2025

Families of Continuous Random Variables

- **Uniform Random Variable:**
- X is a uniform *(a, b)* random variable if the PDF of X is

$$f_X(x) = \begin{cases} {}^{1}\!/_{b-a} & a \le x < b \\ 0 & otherwise \end{cases}$$

- where the two parameters are $b > a$.
- Find CDF, Expected value and variance of the uniform random variable?

Families of Continuous Random Variables

- Example: The phase angle, Θ, of the signal (continuous variable) at the input to a modem is uniformly distributed between 0 and 2π radians. Find the CDF, the expected value, and the variance of Θ?

Families of Continuous Random Variables

- **Exponential Random Variable:**
- X is an exponential (λ) random variable if the PDF of X is

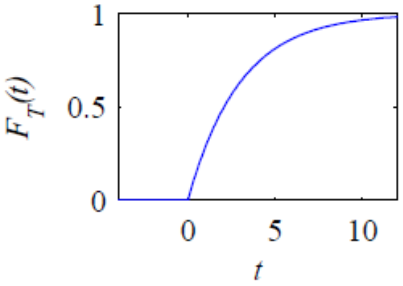
$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & otherwise \end{cases}$$

- where the parameter λ > 0.

- The probability that a telephone call lasts no more than t minutes is often modeled as an exponential CDF.
- What is the PDF of the duration in minutes of

a telephone conversation? What is the probability that

a conversation will last between 2 and 4 minutes?



$$F_T(t) = \begin{cases} 1 - e^{-t/3} & t \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Families of Continuous Random Variables

- What is the expected value and variance in pervious example?

Families of Continuous Random Variables

- If X is an exponential (λ) random variable,
- (a) $F_X(x) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0 & otherwise \end{cases}$
- (b) $E[X] = 1/\lambda$.
- (c) $Var[X] = 1/\lambda^2$.

- **Gaussian Random Variables:**
- *Bell-shaped curves* appear in many applications of probability theory. The probability models in these applications are members of the family of *Gaussian random variables*. Gaussian random is very important variables specially in models of practical phenomena. Because they occur so frequently in practice, Gaussian random variables are sometimes referred to as *normal random variables*.

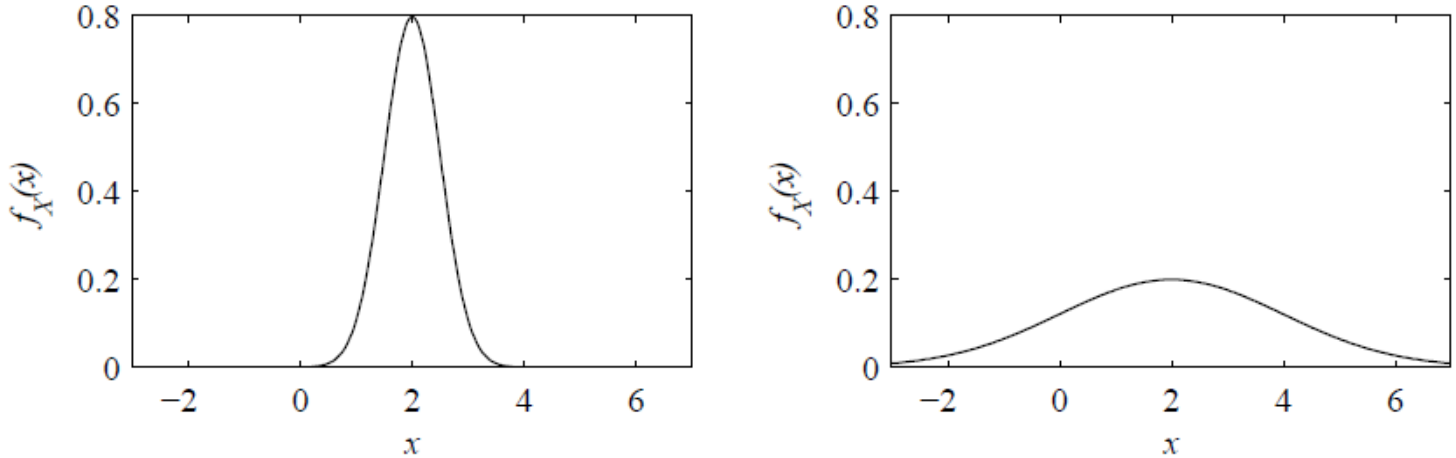
Families of Continuous Random Variables

- **Gaussian Random Variable**
- X is a Gaussian (μ, σ) random variable if the PDF of X is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2 / 2\sigma^2}$$

- where the parameter μ can be any real number and the parameter σ > 0.
- The graph of f X (x) has a bell shape, where the center of the bell is x = μ and σ reflects the width of the bell. If σ is small, the bell is narrow, with a high, pointy peak. If σ is large, the bell is wide, with a low, flat peak.
- If X is a Gaussian (μ, σ) random variable,
- $E\,[X] \;=\; \mu$
- $Var\,[X] \;= \sigma^2$

Families of Continuous Random Variables



(a) $\mu = 2, \sigma = 1/2$ **(b)** $\mu = 2, \sigma = 2$

Families of Continuous Random Variables

- Theorem 1: If X is Gaussian (μ, σ), $Y = aX + b$ is Gaussian $(a\mu + b, a\sigma)$.

- **Definition:** *Standard Normal Random Variable*
- The standard normal random variable is the Gaussian (0, 1) random variable.

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

- **Standard Normal CDF**
- The CDF of the standard normal random variable Z is

$$F_Z(z) = \Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-u^2/2}\, d_u$$

z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$
0.00	0.5000	0.50	0.6915	1.00	0.8413	1.50	0.9332	2.00	0.97725	2.50	0.99379
0.01	0.5040	0.51	0.6950	1.01	0.8438	1.51	0.9345	2.01	0.97778	2.51	0.99396
0.02	0.5080	0.52	0.6985	1.02	0.8461	1.52	0.9357	2.02	0.97831	2.52	0.99413
0.03	0.5120	0.53	0.7019	1.03	0.8485	1.53	0.9370	2.03	0.97882	2.53	0.99430
0.04	0.5160	0.54	0.7054	1.04	0.8508	1.54	0.9382	2.04	0.97932	2.54	0.99446
0.05	0.5199	0.55	0.7088	1.05	0.8531	1.55	0.9394	2.05	0.97982	2.55	0.99461
0.06	0.5239	0.56	0.7123	1.06	0.8554	1.56	0.9406	2.06	0.98030	2.56	0.99477
0.07	0.5279	0.57	0.7157	1.07	0.8577	1.57	0.9418	2.07	0.98077	2.57	0.99492
0.08	0.5319	0.58	0.7190	1.08	0.8599	1.58	0.9429	2.08	0.98124	2.58	0.99506
0.09	0.5359	0.59	0.7224	1.09	0.8621	1.59	0.9441	2.09	0.98169	2.59	0.99520
0.10	0.5398	0.60	0.7257	1.10	0.8643	1.60	0.9452	2.10	0.98214	2.60	0.99534
0.11	0.5438	0.61	0.7291	1.11	0.8665	1.61	0.9463	2.11	0.98257	2.61	0.99547
0.12	0.5478	0.62	0.7324	1.12	0.8686	1.62	0.9474	2.12	0.98300	2.62	0.99560
0.13	0.5517	0.63	0.7357	1.13	0.8708	1.63	0.9484	2.13	0.98341	2.63	0.99573
0.14	0.5557	0.64	0.7389	1.14	0.8729	1.64	0.9495	2.14	0.98382	2.64	0.99585
0.15	0.5596	0.65	0.7422	1.15	0.8749	1.65	0.9505	2.15	0.98422	2.65	0.99598
0.16	0.5636	0.66	0.7454	1.16	0.8770	1.66	0.9515	2.16	0.98461	2.66	0.99609
0.17	0.5675	0.67	0.7486	1.17	0.8790	1.67	0.9525	2.17	0.98500	2.67	0.99621
0.18	0.5714	0.68	0.7517	1.18	0.8810	1.68	0.9535	2.18	0.98537	2.68	0.99632
0.19	0.5753	0.69	0.7549	1.19	0.8830	1.69	0.9545	2.19	0.98574	2.69	0.99643
0.20	0.5793	0.70	0.7580	1.20	0.8849	1.70	0.9554	2.20	0.98610	2.70	0.99653
0.21	0.5832	0.71	0.7611	1.21	0.8869	1.71	0.9564	2.21	0.98645	2.71	0.99664
0.22	0.5871	0.72	0.7642	1.22	0.8888	1.72	0.9573	2.22	0.98679	2.72	0.99674
0.23	0.5910	0.73	0.7673	1.23	0.8907	1.73	0.9582	2.23	0.98713	2.73	0.99683
0.24	0.5948	0.74	0.7704	1.24	0.8925	1.74	0.9591	2.24	0.98745	2.74	0.99693
0.25	0.5987	0.75	0.7734	1.25	0.8944	1.75	0.9599	2.25	0.98778	2.75	0.99702
0.26	0.6026	0.76	0.7764	1.26	0.8962	1.76	0.9608	2.26	0.98809	2.76	0.99711
0.27	0.6064	0.77	0.7794	1.27	0.8980	1.77	0.9616	2.27	0.98840	2.77	0.99720
0.28	0.6103	0.78	0.7823	1.28	0.8997	1.78	0.9625	2.28	0.98870	2.78	0.99728
0.29	0.6141	0.79	0.7852	1.29	0.9015	1.79	0.9633	2.29	0.98899	2.79	0.99736
0.30	0.6179	0.80	0.7881	1.30	0.9032	1.80	0.9641	2.30	0.98928	2.80	0.99744
0.31	0.6217	0.81	0.7910	1.31	0.9049	1.81	0.9649	2.31	0.98956	2.81	0.99752
0.32	0.6255	0.82	0.7939	1.32	0.9066	1.82	0.9656	2.32	0.98983	2.82	0.99760
0.33	0.6293	0.83	0.7967	1.33	0.9082	1.83	0.9664	2.33	0.99010	2.83	0.99767
0.34	0.6331	0.84	0.7995	1.34	0.9099	1.84	0.9671	2.34	0.99036	2.84	0.99774
0.35	0.6368	0.85	0.8023	1.35	0.9115	1.85	0.9678	2.35	0.99061	2.85	0.99781
0.36	0.6406	0.86	0.8051	1.36	0.9131	1.86	0.9686	2.36	0.99086	2.86	0.99788
0.37	0.6443	0.87	0.8078	1.37	0.9147	1.87	0.9693	2.37	0.99111	2.87	0.99795
0.38	0.6480	0.88	0.8106	1.38	0.9162	1.88	0.9699	2.38	0.99134	2.88	0.99801
0.39	0.6517	0.89	0.8133	1.39	0.9177	1.89	0.9706	2.39	0.99158	2.89	0.99807
0.40	0.6554	0.90	0.8159	1.40	0.9192	1.90	0.9713	2.40	0.99180	2.90	0.99813
0.41	0.6591	0.91	0.8186	1.41	0.9207	1.91	0.9719	2.41	0.99202	2.91	0.99819
0.42	0.6628	0.92	0.8212	1.42	0.9222	1.92	0.9726	2.42	0.99224	2.92	0.99825
0.43	0.6664	0.93	0.8238	1.43	0.9236	1.93	0.9732	2.43	0.99245	2.93	0.99831
0.44	0.6700	0.94	0.8264	1.44	0.9251	1.94	0.9738	2.44	0.99266	2.94	0.99836
0.45	0.6736	0.95	0.8289	1.45	0.9265	1.95	0.9744	2.45	0.99286	2.95	0.99841
0.46	0.6772	0.96	0.8315	1.46	0.9279	1.96	0.9750	2.46	0.99305	2.96	0.99846
0.47	0.6808	0.97	0.8340	1.47	0.9292	1.97	0.9756	2.47	0.99324	2.97	0.99851
0.48	0.6844	0.98	0.8365	1.48	0.9306	1.98	0.9761	2.48	0.99343	2.98	0.99856
0.49	0.6879	0.99	0.8389	1.49	0.9319	1.99	0.9767	2.49	0.99361	2.99	0.99861

Families of Continuous Random Variables

- Given a table of values of $\Phi(z)$, we use the following theorem to find probabilities of a Gaussian random variable with parameters μ and σ.
- ***Theorem:*** *If X is a Gaussian (μ, σ) random variable, the CDF of X is*

$$F_X(x) = \Phi(\frac{x-\mu}{\sigma})$$

- *The probability that X is in the interval (a, b] is*

$$P[a < X \leq b] = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$

- In using this theorem, we transform values of a Gaussian random variable, *X*, to equivalent values of the standard normal random variable, *Z*. For a sample value x of the random variable X, the corresponding sample value of Z is

$$z = \frac{x-\mu}{\sigma}$$

Families of Continuous Random Variables

- Example 1: Suppose your score on a test is $x = 46$, a sample value of the Gaussian (61, 10) random variable. Express your test score as a sample value of the standard normal random variable, Z.

- Example 2: If X is the Gaussian (61, 10) random variable, what is $P[X \leq 46]$?

- Example 3: if X is a Gaussian random variable with $\mu = 61$ and $\sigma = 10$, what is $P[51 < X \leq 71]$?

Families of Continuous Random Variables

- Example: X is the Gaussian (0, 1) random variable and Y is the Gaussian (0, 2) random variable.
- (1) Sketch the PDFs $f_X(x)$ and $f_Y(y)$ on the same axes.
- (2) What is $P[-1 < X \leq 1]$?
- (3) What is $P[-1 < Y \leq 1]$?
- (4) What is $P[Y > 3.5]$?

Families of Continuous Random Variables

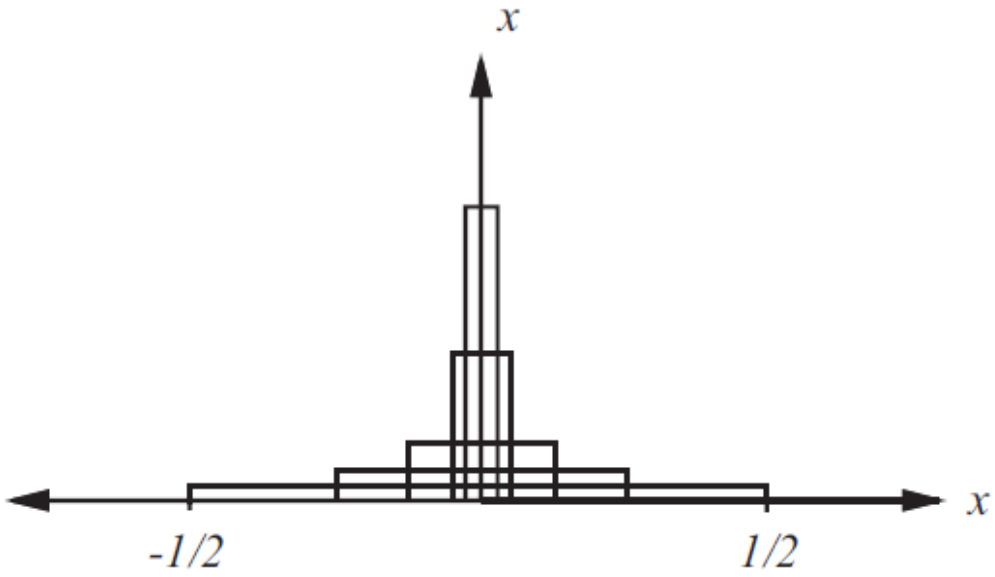
- **Delta Functions**
- **Unit Impulse (Delta) function:**
- Suppose the following function:

$$d_\epsilon(x) = \begin{cases} 1/\epsilon & \frac{-\epsilon}{2} \le x \le \frac{\epsilon}{2} \\ 0 & otherwise \end{cases}$$

- *The **unit impulse function** is:*

$$\delta(x) = \lim_{\epsilon \to 0} d_\epsilon(x)$$

Families of Continuous Random Variables



- For each ϵ the area under the curve of $d_\epsilon(x)$ equals 1.

Families of Continuous Random Variables

- **Theorem:** For any continuous function g(x),

$$\int_{-\infty}^{+\infty} g(x)\delta(x - x_0)dx = g(x_0)$$

- **Unit Step Function**

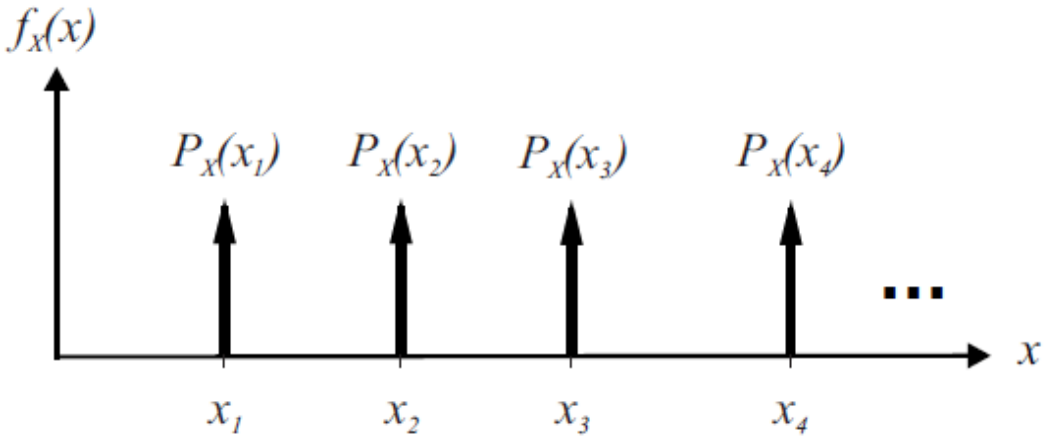
$$u(x) = \begin{cases} 0 & x < 0 \\ 1 & x \geq 0 \end{cases}$$

- The relation between unit impulse and unit step functions:

$$\delta(x) = \frac{du(x)}{dx}$$

Families of Continuous Random Variables

- Lets assume that we have a PDF function as below:



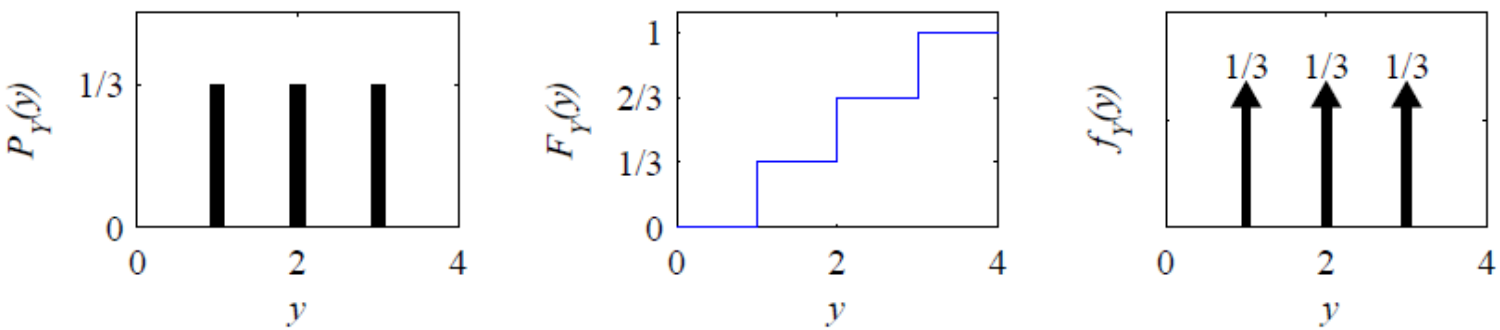
$$f_X(x) = \sum_{all\ x} P_X(x_i)\delta(x - x_i)$$

- When the PDF includes delta functions of the form $\delta(x - x_i)$, we say there is an impulse at x_i .

Families of Continuous Random Variables

- When we graph a PDF $f_X(x)$ that contains an impulse at x_i , we draw a vertical arrow labeled by the constant that **multiplies** the impulse. We draw each arrow representing an impulse at the same height because the PDF is always **infinite** at each such point.
- To calculate the expected value:

$$E[X] = \int_{-\infty}^{+\infty} x \left(\sum_{all\ x} P_X(x_i)\delta(x - x_i) \right) dx$$



Families of Continuous Random Variables

- Example 1: Suppose Y takes on the values 1, 2, 3 with equal probability. The PMF and the corresponding CDF of Y are (figure in previous page)

$$P_Y(y) = \begin{cases} \frac{1}{3} & y = 1, 2, 3 \\ 0 & otherwise \end{cases}$$

- Using the unit step function u(y), we can write CDF $F_Y(y)$ and using impulse function we can find the PDF as follow:

- Consequently We see that the discrete random variable Y can be represented graphically either by a PMF or CDF or PDF

Families of Continuous Random Variables

- Example 2: The cumulative distribution function of random variable X is
- $F_X(x) = \begin{cases} 0 & x < -1 \\ \frac{x+1}{4} & -1 \leq x < 1 \\ 1 & x \geq 1 \end{cases}$
- Sketch the CDF and find the following:
- (1) P[X ≤ 1]
- (2) P[X < 1]
- (3) P[X = 1]
- (4) the PDF $f_X\ (x)$

Families of Continuous Random Variables

- Example 3: Observe someone dialing a telephone and record the duration of the call. In a simple model of the experiment, 1/3 of the calls never begin either because no one answers or the line is busy. The duration of these calls is 0 minutes. Otherwise, with probability 2/3, a call duration is uniformly distributed between 0 and 3 minutes. Let Y denote the call duration. Find the CDF , the PDF , and the expected value E[Y]?